RUSHHOUR3 · PROJECT 75

JetRacer Autonomous Driving and Delivery

Teaching a real car to drive itself
with reinforcement learning

Augustus Amedume **24983037**
SAC-VAE on the real car · VAE · dashboard

Seth Yap Wing Shen **25965486**
Imitation learning · 2D-sim DQN · OpenCV

Jia Lu **25404498**
Image-based DQN in DonkeySim



43008 Reinforcement Learning · Assessment 3 · Part F

Delivery still depends on human drivers

Jia Lu

THE PROBLEM TODAY

Driver fatigue

Long shifts reduce alertness

Operating costs

Every vehicle needs a paid driver

Inconsistent performance

Timing and quality vary between drivers

OUR PROOF OF CONCEPT

A JetRacer as a small autonomous delivery vehicle

- Follows its lane using only the onboard camera
- Steers through the corners on its own
- Stops at pickup and drop-off zones (final goal)

Our question: can a real car learn reliable lane following from camera images with reinforcement learning?

A real track, a real car, a noisy camera

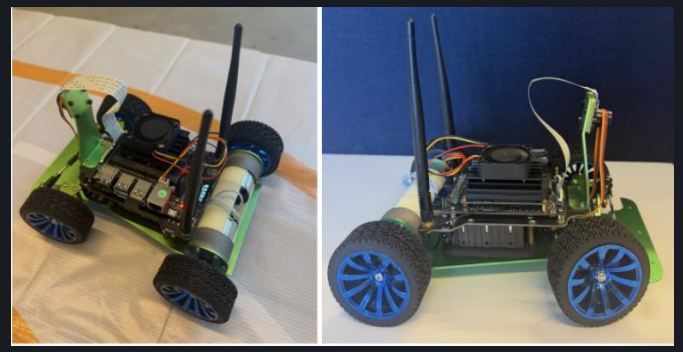
Seth Yap Wing Shen

TRACK



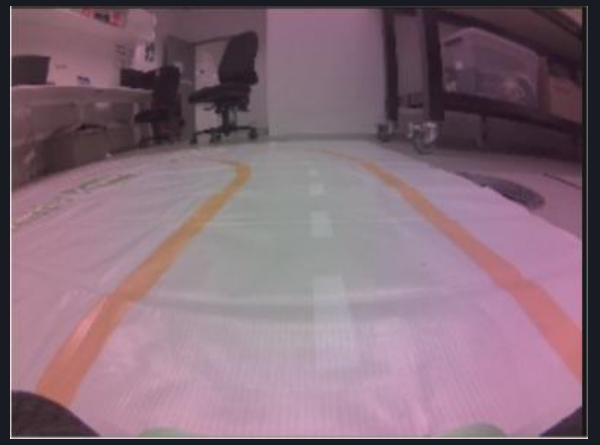
- Oval, about 2.65 m × 1.70 m
- Lane 0.26 m wide between orange lines
- Dashed white centre line

JETRACER



- Jetson Nano runs it all on board
- Front camera, 320 × 240 colour
- Steering servo + drive motor

CAMERA VIEW



- Pink colour cast at the edges
- Room and people above the track
- Datasets: 849 frames (imitation), 6,432 + 15,809 (VAE)

We also built two simulators: a 2D Gymnasium track (slide 6) and a DonkeySim copy of the lab track (slide 7).

Four approaches, one goal

Seth



APPROACH 1

Imitation learning

ResNet18 copies a human driver

Owner: **Seth**

RESULT

Val. MAE 0.086, but missed corners on the car



APPROACH 2

DQN + OpenCV

Trained in a 2D simulator, run through lane detection

Owner: **Seth**

RESULT

4.6 laps in sim, unreliable on the car



APPROACH 3

Image-based DQN

Trained on camera images in DonkeySim

Owner: **Jia**

RESULT

11/12 sim runs, not tested on the real track



SELECTED

APPROACH 4

SAC + VAE

Learns online on the real car

Owner: **Augustus**

RESULT

≈4.5 laps on the real track

Imitation learning: copy a human driver

Seth Yap Wing Shen

Drive by keyboard
teleoperation, 4 sessions

→

Record pairs
camera frame + steering label

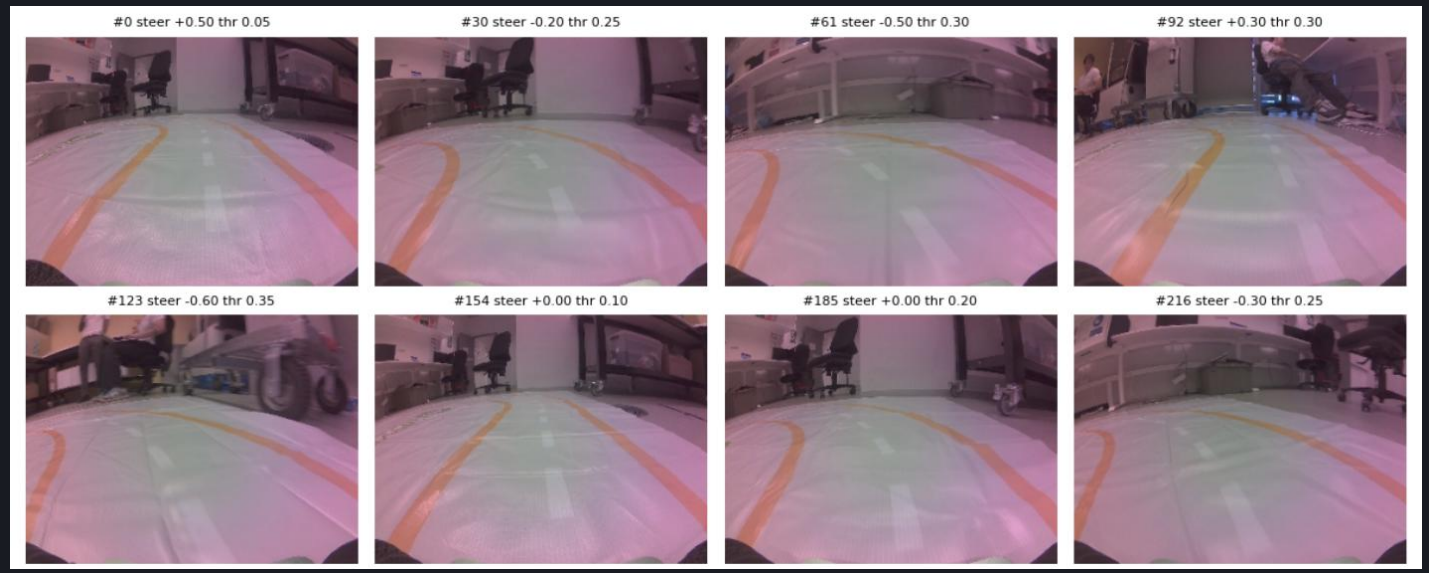
→

Train ResNet18
ImageNet weights, 1 steering output

→

Steer live
predicted steering, fixed throttle

TRAINING PAIRS: CAMERA FRAME + STEERING LABEL



0.086

validation MAE (steering)

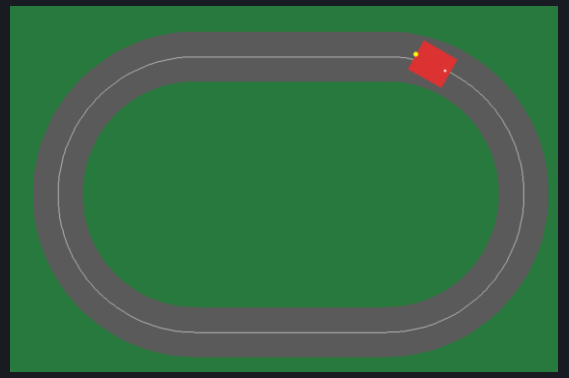
- 756 of 849 frames kept
- 608 train / 148 val, split by time
- L1 loss · AdamW · 20 epochs

On the car: fine on straights, missed corners

2D-sim DQN, deployed through OpenCV

Seth Yap Wing Shen

1 · TRAIN IN A 2D GYMNASIUM SIMULATOR



State

lane offset, heading error, last steering

Actions

left · straight · right

Reward

+ progress near the centre

– leaving the lane

Training

SB3 DQN, 20 min (5.68 M steps)

2 · DEPLOY WITH OPENCV LANE DETECTION



Orange boundaries → lane offset + heading → DQN → steering

4.6 laps on average in simulation

0% crashes over 100 greedy test episodes

Unreliable on the car

Lost a boundary on corners; sensitive to noise and delays

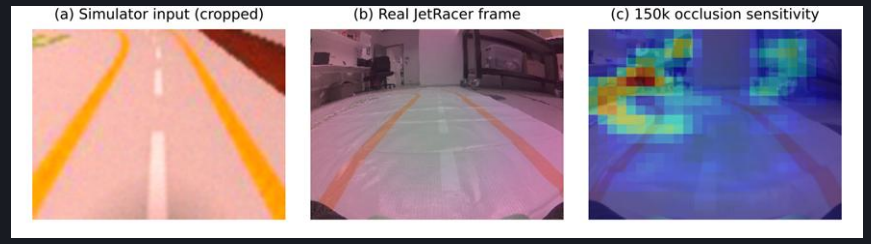
Image-based DQN in DonkeySim

Jia Lu

UNITY COPY OF THE LAB TRACK



SIMULATOR VS REAL CAMERA



(c) shows the first model looked mostly at the room (red)

1 **Lane following**
from scratch · 150k steps

2 **Crop + random colours**
top 45% removed · 125k

3 **Mirrored runs**
resized track · 750k

Model	CCW	CCW + rand.	CW (mirror)	CW + rand.	Total
Start*	2/2	–	**0/2**	–	2/4
Round 1	3/3	3/3	**0/3**	**0/3**	6/12
Round 2	3/3	3/3	3/3	**1/3**	10/12
Round 3 (final)	3/3	3/3	3/3	2/3	**11/12**

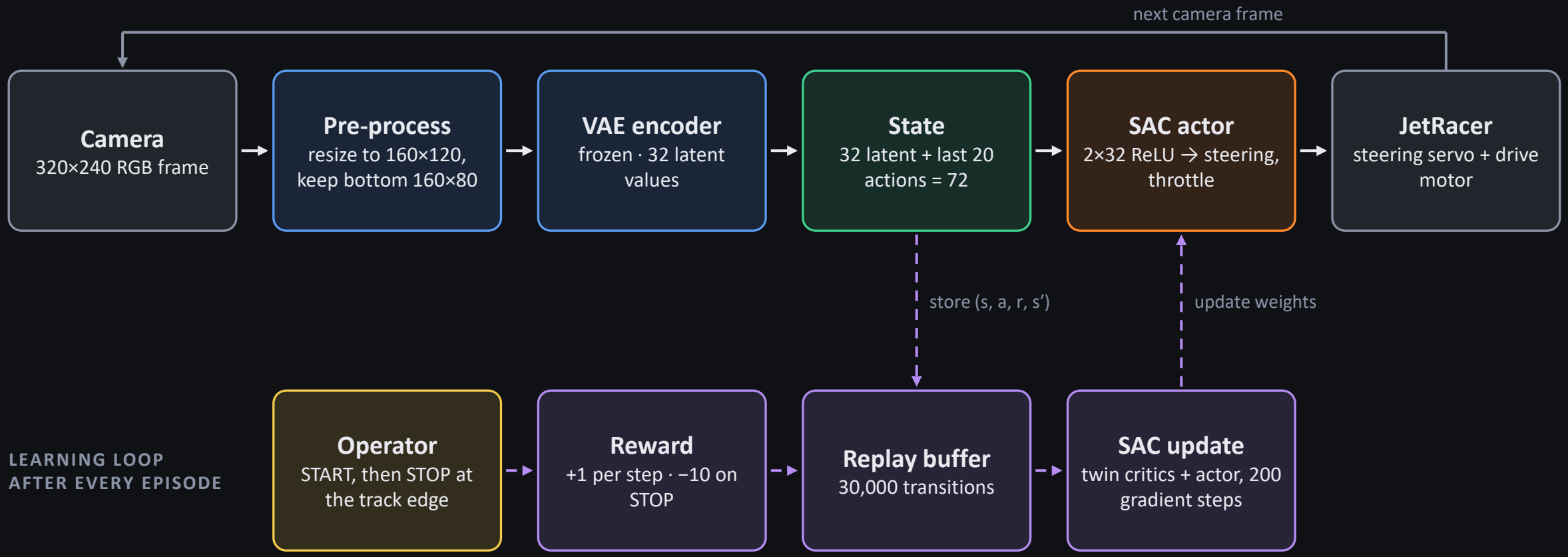
Runs reaching 1,000 steps on the resized track. *Start model: 2 runs per condition, 500-step limit.

⚠ On real frames its steering agreed better with the human after cropping + colours, but it never drove the real track (servo off-centre, power loss).

SAC-VAE: separate seeing from deciding

Augustus Amedume

DRIVING LOOP · EVERY STEP (≈22–35 Hz)



Seeing: VAE, trained offline, frozen

Deciding: SAC actor, learned on the car

Learning: after every episode

Human-in-the-loop training on the car

Augustus Amedume

State	72 values: VAE latent (32) + last 20 actions (40)
Action	Steering in [−0.8, 0.8]; throttle 0.3–0.4 forward; steering change ≤ 0.56 per step
Reward	1 + 0.1 · throttle / 0.4 per step; −10 − 5 · throttle (normalised) on STOP
Discount	$\gamma = 0.99$
Networks	Actor + twin critics, 2 × 32 ReLU · lr 3×10^{-4} · batch 64 · buffer 30,000

EACH EPISODE



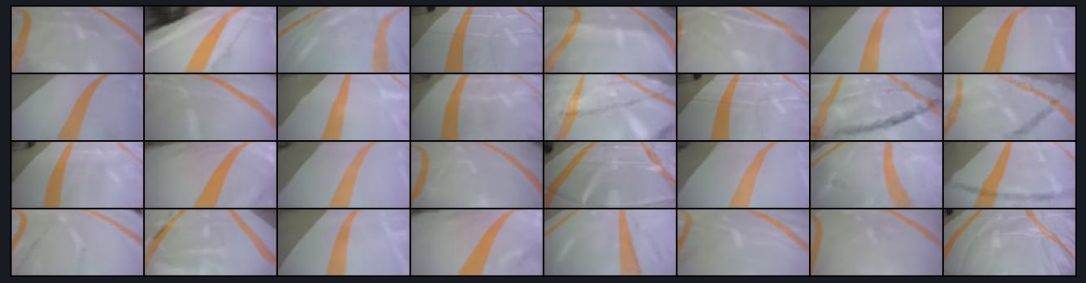
Each session starts with 300 random warm-up steps. No simulator: all learning ran on the Jetson Nano.

Seeing the track: VAE v1 vs v2

Augustus Amedume

VAE V1 · 6,432 IMAGES · ANTI-CLOCKWISE ONLY

In the training notebook: looks fine



Live, on a right turn: noise



Input crop



Reconstruction · error 485,386

VAE V2 · 15,809 IMAGES · BOTH DIRECTIONS + GLARE

- Recorded clockwise and anti-clockwise
- Different light and glare conditions
- Same 160 × 80 crop; trained for 62 epochs

Live frame (dashboard): track recovered



Input crop



Reconstruction

The driving results we report used VAE v1. VAE v2 and a corrected steering configuration were ready, but hardware faults stopped retraining.

Results: 87 → 343 steps per episode

Augustus Amedume



3,130

longest episode: over 2 minutes driving alone

≈26,500

steps · 200 episodes · 6 sessions

≈4.5 laps

deterministic test runs: 37.9 s and 30.9 s anti-clockwise

Clockwise: failed on right-hand curves

Demo: the policy driving on its own

Augustus Amedume



No human steering

5+ laps in 83 s

Anti-clockwise

LIVE DASHBOARD · SERVED FROM THE CAR, VIEWED IN ANY BROWSER

JetRacer RushHour3 DEMO · live

CAMERA (WHAT THE VAE SEES)

VAE RECONSTRUCTION

RECONSTRUCTION LOSS

1,934

STEERING

-0.37 left

left right

THROTTLE

0.35 forward

0 0.40

Limitations and lessons

Jia Lu

Sim-to-real gap

Both DQNs were reliable in simulation, where states are perfect; the car must estimate them from a noisy camera.

Perception

OpenCV lost a boundary on corners; the camera has a pink colour cast; VAE v1 failed on unseen right turns.

Steering hardware

Servo centre at +0.3: the car turned further left (1.1) than right (0.5). A failing servo later cut the power.

One direction only

SAC was trained anti-clockwise only, so it never learned to drive clockwise.

Survival-only reward

The reward pays for staying on the track, not for staying centred in the lane.

Delivery unfinished

Pickup and drop-off detection and automatic stopping were not completed.

Conclusion and next steps

Jia Lu

WHAT WE FOUND



SAC-VAE gave the best real-world result: multiple laps after learning directly on the car



Simulation-trained DQNs did well in simulation but did not transfer reliably



Perception, calibration and training conditions matter as much as the algorithm

NEXT STEPS

1. Retrain SAC in both directions with VAE v2 and the corrected steering
2. Evaluate over repeated trials, not single runs
3. Add pickup and drop-off detection and stopping

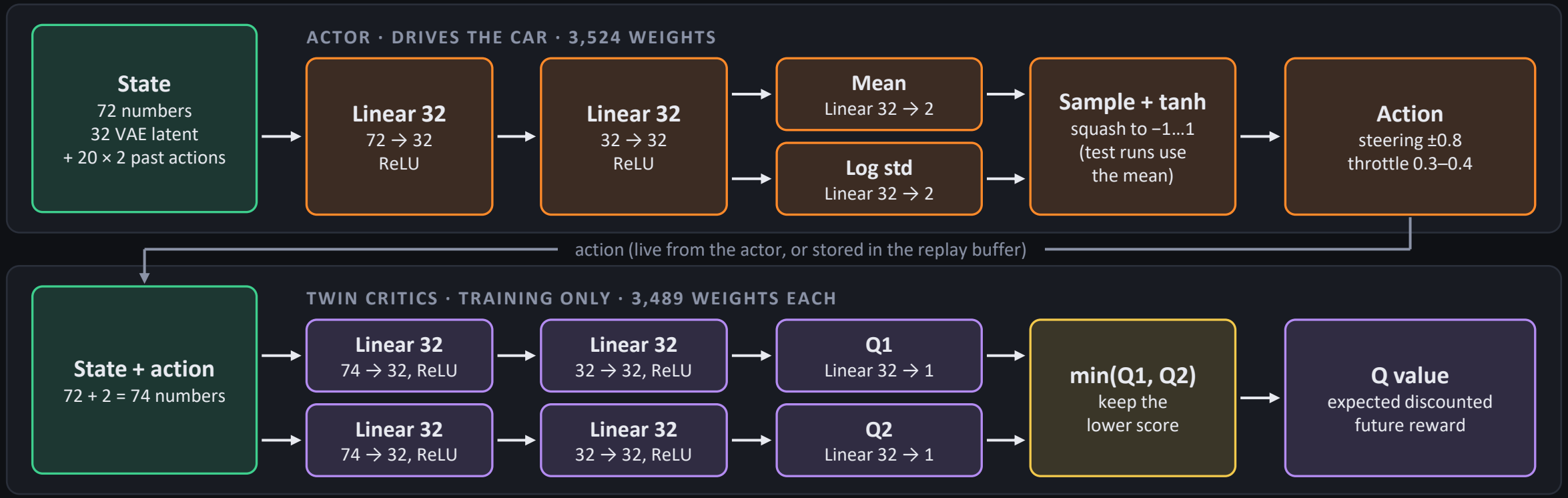
Thank you.

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github.com/dumes21/AutonomousDriving-

Backup: SAC's actor and twin critics

Backup · Augustus



AFTER EVERY EPISODE: 200 UPDATES · BATCHES OF 64 · 30,000-STEP REPLAY BUFFER

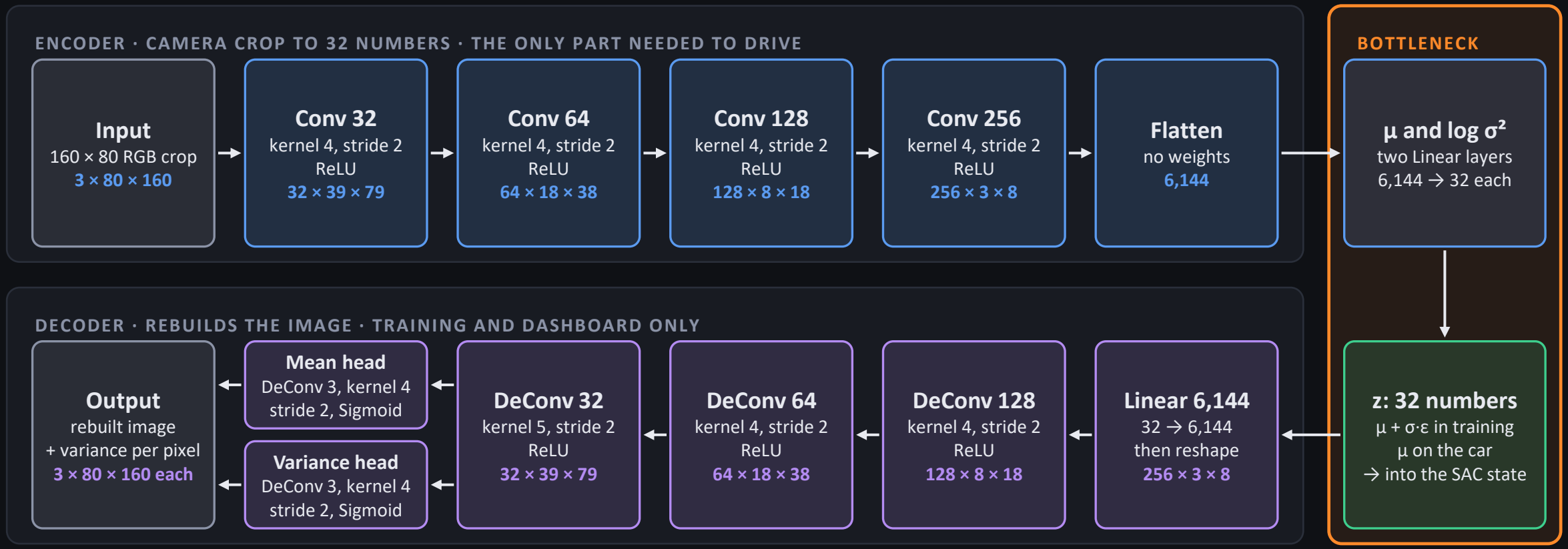
Critics
Pulled toward $r + \gamma (\min Q'(s', a') - \alpha \log \pi(a'|s'))$, or just r at a STOP

Actor
Minimises $\alpha \log \pi(a|s) - \min(Q1, Q2)$: aim for a high Q, stay a bit random

Target critics and α
Q' are slow copies of both critics ($\tau = 0.02$). α tunes itself to keep entropy near −2

Backup: the VAE layer by layer

Backup · Augustus



Loss

½ Σ [(x − mean)² / var + log(2π · var)] + KL

Why a variance head

It predicts how sure it is of each pixel, which is why the loss can go negative

Size

1,996,038 weights. VAE v1 and v2 share this network. Shapes: channels × height × width